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MILITARY SPECIFICATION

SHELTER, JOINING CORRIDOR FOR

This specification is approved for use by the United States Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Joining Corridor used with the Marine Corps Expeditionary Shelter System.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-1, Washington D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5411

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SPECIFICATIONS

MILITARY

MIL-C-5541	Chemical Conversion Coating on Aluminum Alloys
MIL-H-21040	Honeycomb Materials, Water Migration Resistant Type, Structural Paper Base
MIL-S-29409	Shelter Assembly, Knockdown
MIL-S-29410	Shelter Assembly, Rigid
MIL-S-29411	Shelter Assembly, 10-ft and 20-ft EMI

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-401	Sandwich Construction and Core Materials, General Test Methods
MIL-STD-810	Environmental Test Methods and Engineering Guidlines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-907	Engineering and Design Criteria for Shelters, Expandable and Nonexpandable
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

MCO P4790.1	MIMMS Introduction Manual

DRAWINGS

MARINE CORPS

DL80A5022A0004	Corridor Assembly, Joining

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

INTERNATIONAL STANDARD

ISO 668 Dimensions and Ratings

(Application for copies should be addressed to International Organization for Standardization, P.O. Box 56, 1211 Geneva 20, Switzerland.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-864-82 Surface Preparation of Aluminum Alloys to be Adhesively Bonded in Honeycomb Sandwich Panels

ASTM E-865-82 Structural Film Adhesives for Honeycomb Sandwich Panels

ASTM E-866-82 Corrosion-Inhibiting Adhesive Primer for Aluminum Alloys to be Adhesively Bonded in Honeycomb Sandwich Panels

ASTM E-874-82 Adhesive Bonding of Aluminum Facings to Nonmetallic Honeycomb Core for Shelter Panels

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supercede

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applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article

3.1.1 Sandwich panel, large. One 4-ft (1.219-m) by 8-ft (2.438-m) aluminum face honeycomb sandwich panel fabricated in accordance with the requirements specified in 3.3.1 and Drawing No. 80A5022A0071 shall be subjected to the panel shear test in 4.8.1 and shall be subsequently cut into smaller specimens for other panel testing (see 4.8.2 and 4.8.5).

3.1.2 Assemblies. When specified in the contract or purchase order, the joining corridor shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. Materials shall conform to the requirements specified herein and in Data List DL80A5022A0004.

3.3 Design and construction

3.3.1 Sandwich panels. Sandwich panels shall be fabricated and inspected in accordance with the appropriate materials and process specifications listed in Data List DL80A5022A0004 and as follows:

3.3.1.1 Sandwich panels and material. The following material requirements are the minimum necessary to insure the joining corridor fully meets all requirements stated herein.

3.3.1.1.1 Sandwich panel shear strength. The adhesive bond between the panel facings and core shall be sufficient to allow the core to work to its failing shear stress. The composite sandwich panel shear strength at normal temperature (75° ± 5°F) shall be 80 psi (551,584 N/m^2) in the T_L direction parallel to the core ribbons. Testing shall be in accordance with 4.8.1.

3.3.1.1.2 Impact resistance. A 24-inch (in) (0.610-m) square sample of all exterior shelter panel(s) shall withstand a blow from a 70-pound (lb) (31.8-kilogram (kg)) steel cylinder (3 in (76.2 millimeter (mm)) in diameter with a hemispherical end), dropped from a height of 30 in (0.762 m) (16 in (0.406m) for 2.5# density honeycomb), impacting the hemispherical end without rupture to either skin and no delamination between skins and core outside a 3-in (76.2-mm) radius from the center of impact. Localized crushing of the core is allowable.
Testing shall be in accordance with 4.8.2.

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3.3.1.2 Surface preparation of aluminum alloys to be adhesively bonded in honeycomb shelter panels. Preparation of clean uniform surfaces of aluminum alloys suitable for formation of durable adhesive bonds to nonmetallic honeycomb materials in the manufacture of sandwich panels for the joining corridor shall be in compliance with ASTM E-864-82. Honeycomb core material must comply with MIL-H-21040.

3.3.1.3 Structural film adhesives for honeycomb sandwich panels. Film adhesives used for bonding of honeycomb sandwich panels used in the shelter assembly shall be in accordance with ASTM E-865-82 and shall be suitable for forming bods which can withstand long exposures to temperatures from -70°F (-56.7° Celsius (C)) to +160°F (+71.1°C) and also withstand the combinations of stress, temperature, and relative humidity expected to be encountered in service. The adhesives shall be used for bonding aluminum alloy facings to nonmetallic core, inserts, edge attachments and other components of a sandwich panel. The adhesives may be used for new production or depot repair. Honeycomb core materials will meet requirements of MIL-H-21040. The adhesive shall meet the requirements of the test specified in 4.8.10.

3.3.1.4 Corrosion-inhibiting adhesive primer for aluminum alloys to be adhesively bonded in honeycomb shelter panels. Primer shall be a sprayable, pigmented liquid primer for use on aluminum alloys which are to be adhesively bonded in the fabrication of honeycomb sandwich panels for the shelter assembly. When applied to a properly cleaned surface of aluminum alloy, the primer will impart corrosion resistance and form a surface suitable for structural bonding using adhesives complying with ASTM E-865-82 and for coating with shelter paint finishes. Corrosive-inhibiting adhesive primer shall comply with standards contained in ASTM E-866-82. All surfaces to be bonded will comply with this standard or meet performance standards using alternative chemical treatment processes to be specifically approved by the contracting agency.

3.3.1.5 Adhesive bonding of aluminum facings to nonmetallic honeycomb core for shelter panels. Materials, processes and quality controls to be used in the manufacture of adhesive-bond, aluminum-faced, nonmetallic, honeycomb-core sandwich panels for the joining corridor shall comply with standards contained in ASTM E-874-82.

3.3.1.6 Core splice adhesive for honeycomb sandwich shelter panels. The adhesive shall be suitable for forming bonds which can withstand long combinations of stress, temperature, and relative humidity expected to be encountered in service. The adhesive shall be used for bonding core segments, bonding core segments to edge members, and integral members in a honeycomb sandwich panel. The adhesive shall not allow the transfer of water from one part of the panel to another either through or along the line of the adhesive.

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3.3.2 Fastenings and joints. Welding, riveting and sealing of joints shall comply with the applicable standards and drawings in Data List DL80A5022A0004.

3.3.3 Panel flatness. Panel surfaces shall not be cupped or bowed in excess of 0.125 in (3.175 mm) when measured with a 48-in (1.219-m) long straightedge. Panels shall be tested in accordance with 4.8.3.

3.3.4 Interchangeability. Like demountable hardware and components of the joining corridor shall be interchangeable.

3.3.5 Ease of operation. The demountable side panels shall permit repetitive installation and removal without binding or other loss of serviceability. Doors shall operate smoothly, maintaining adequate sealing at all times. The entire assembly shall be capable of multiple assembly and disassembly operations without degradation of components.

3.3.6 Fire resistance. The joining corridor shall be fire resistant and shall meet the test requirements of 4.8.5.

3.3.7 Complexing. The joining corridor shall be capable of complexing two to four shelter assemblies by connection to each of the four sides of the corridor and the door of the shelter. No additional material is required to adequately seal the joining corridor to the shelter assembly(ies).

3.3.8 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and punching shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.3.9 Facing material. Panel facings shall be one piece seamless aluminum alloy skins of a material and thickness specified on the applicable drawings. All edges of aluminum faced honeycomb panels shall be completely sealed to prevent water intrusion between the laminates.

3.3.10 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled. All bolts and rivet connectors shall have the burrs removed. Washers, lockwashers, or locknuts shall be provided where necessary and all bolts, nuts and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter rivet, concentric with the rivet holes, and in full contact with the surface of the members.

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3.3.11 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillets shall be as specified on the applicable drawings.

3.3.12 Castings. Castings shall be sound and free from patching, misplaced coring, warping or defects which might render the castings unsound for use.

3.3.13 Seals. Seals shall be installed so that the fit within the retaining track is snug and continuous contact is maintained with mating seal faces. Mitered end junctions of vertical and horizontal seal runs shall be accurate, and gaps shall be sealed as specified on the applicable drawings.

3.4 Maintainability. The joining corridor has been designed so that as much of the maintenance, as possible, may be performed as far forward (preferably by the using unit), as feasible, within the Marine Corps maintenance structure, in accordance with MCO P4790.1.

3.5 Environmental requirements

3.5.1 Rainfall. The joining corridor when erected and with doors closed and removable panels installed shall withstand rainfall up to and including 5 in per hour (in/hr) (127 millimeters per hour (mm/hr)) without leakage at any location. Testing shall be in accordance with 4.8.7.1.

3.5.2 Temperature-humidity. In storage, the joining corridor shall be capable of withstanding exposure to temperatures of -70°F (-56.7°C) to 160°F (71.1°C). In transit, the joining corridor shall be capable of withstanding exposure to temperatures of -65°F (-53.9°C) to 160°F (71.1°C) with personnel access at low end of range. Operational temperature of joining corridor shall be -65°F (-53.9°C) to 125°F (51.7°C) plus solar load (see 3.5.8). The joining corridor shall withstand daily exposure of up to 97 percent relative humidity for 20 hours and exposure of 100 percent relative humidity (with condensation) for 4 hours. Testing shall be in accordance with 4.8.7.2 and 4.8.7.3.

3.5.2.1 Thermal shock. The joining corridor shall withstand sudden temperature changes from 160°F (71.1°C) to -70°F (-56.7°C) to 160°F (71.1°C) without separation, delamination, cracks or degradation of physical properties. Testing shall be in accordance with 4.8.7.4.

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3.5.2.2 Heat transfer. The joining corridor shall maintain an overall average heat transfer coefficient less than or equal to 0.35 British thermal unit (BTU)/hr-Ft ^{2}-°F (1.99 W/m ^{2} -°K). Testing shall be in accordance with 4.8.7.5.

3.5.3 Snow and ice. The joining corridor shall be capable of functioning in accordance with the requirements of this specification during periods of snow and ice storms. The corridor shall inhibit the collection of snow and ice so that its primary function is not degraded. The corridor shall be capable of snow loads of up to 40 lbs per square ft (1,915 N/m ^{2}) in addition to its internal load.

3.5.4 Wind. The joining corridor shall be capable of withstanding steady state wind speeds of up to 65 mph (104 km/hr) and gusts up to 120 mph (192 km/hr), static load on the door of 15 lbs per square ft (718 N/m ^{2}) and a dynamic load on the door of 50 lbs (22.7 kg) applied to the midpoint of the locking edge acting in a horizontal plane perpendicular to the front panel of the door in both directions, the force increased by dropping the weight (50 lbs) a distance of 12 in (0.305 m). Testing shall be in accordance with 4.8.7.6. In winds in excess of 55 mph (88 km/hr), ground anchors are required.

3.5.5 Roof load. The joining corridor shall withstand a uniform load of 40 lbs per square foot (1,915 N/m ^{2}) over the entire roof of the corridor without permanent damage and a concentrated load of 660 lbs (300 kg) over an area 24 in x 12 in (0.610 m by 0.305 m) without permanent damage rendering the corridor assembly unusable. Testing shall be in accordance with 4.8.7.7 and 4.8.7.8.

3.5.6 Sand and dust. The joining corridor shall withstand the sand and dust effects on all external moving parts with particle concentrations of 0.000132 pounds per cubic foot (lb/cu-ft ^{3}) (2.19 gm/m ^{3}), with wind velocity of 1,750 ± 250 ft/min (533 ± 76.2 m/min) without degradation. Such particles shall range in size from 0.00294 in (0.150 mm) up to 0.0197 in (1.0 mm) with relative humidity less than 23 percent. Testing shall be in accordance with 4.8.7.9.

3.5.7 Sunshine. The joining corridor shall withstand the effects of ultraviolet radiation (sunshine) without significant degradation or effect on serviceability of the corridor's components or materials during the service life of the corridor, and shall withstand the test requirements of MIL-STD-810, Method 505.1. Testing shall be in accordance with 4.8.7.10.

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3.5.8 Solar load. The joining corridor shall withstand a solar heat load on its roof sufficient to raise the outer skin to at least 200°F (93.3°C) without any evidence of delamination or permanent deformation. Testing shall be in accordance with 4.8.7.11.

3.5.9 Marine environment. The joining corridor shall be fully serviceable when exposed to sea-salt fallout equivalent to 12.5 pounds/acre/year (1.4 gm/m^2/year). All fasteners, jacks, seals, or other hardware shall show no evidence of corrosion or degradation when tested as required by 4.8.7.12.

3.5.10 Drop shock. The joining corridor shall withstand without damage the following drop shocks when in its shipping mode (folded). After each test, there shall be no permanent deformation, buckling, delamination sealer separation or structural failures to any part of the corridor. Doors and covers shall open and close to their full extent without binding.

3.5.10.1 Flat drop. The joining corridor shall withstand a flat drop of 12 in (300 mm) from the ground onto a concrete surface. Testing shall be in accordance with 4.8.8.1.

3.5.10.2 Rotational drop. The joining corridor shall withstand rotational drops, with one edge placed on a 4-in (102-mm) block and the opposite edge raised 12 in (305 mm) and allowed to free fall onto a concrete surface with the 4-in (102-mm) block acting as a pivot. The corridor shall withstand this drop on each bottom edge. Testing shall be in accordance with 4.8.8.2.

3.5.11 Blackout. The joining corridor shall be light-tight with the door closed and complexed to one, two or three shelters. Testing shall be in accordance with 4.8.9.1.

3.5.12 Lifting from top. The joining corridor shall withstand lifting accelerations of up to 2.5 gravities when supported solely by the four lifting eyes in its travel configuration. Testing shall be in accordance with 4.8.6.

3.6 Dimensional characteristics. The exterior envelope of the joining corridor shall be 132 in (3.353 m) long, 86 in (2.184 m) wide and 86.5 in (2.197 m) high. The applicable drawings fully describe all pertinent dimensions and tolerances. Should any dimensions and tolerances result in a component or assembly which will not meet the requirements of this specification, the supplier is responsible for fabrication techniques which will prevent the occurrence of such dimensions and tolerances. Testing shall be in accordance with 4.8.4.

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3.7. Weight. The tare weight of the joining corridor shall be 660 lbs (2,936 Newtons). The actual weight shall be stamped on the nameplate.

3.8 Finish

3.8.1 Surface preparations. All surfaces shall be prepared for painting as specified on the applicable drawings. All aluminum alloy materials shall be protected by a chemical film conforming to MIL-C-5541 even if a final finish is not applied.

3.8.2 Priming. All surfaces to be painted shall be primed as specified on the applicable drawings. Panel edge members and frames to be processed in accordance with ASTM E-864-82 and protected from corrosion by a chemical conversion coating in accordance with MIL-C-5541, Class 3, or a corrosion inhibiting adhesive primer in accordance with ASTM E-866-82.

3.8.3 Final finish. Painted surfaces shall be as specified on the applicable drawings.

3.9 Identification. Each joining corridor shall have the following information provided on a serial plate permanently affixed to the corridor in accordance with MIL-STD-130:

Manufacturer:
Model:
Type:
Serial No.: (As assigned by the Procuring Agency)
Weight:

3.10 Workmanship. All parts, components and assemblies of the joining corridor including casting, forgings, molded parts, stampings, seals and sealing agents, machined surfaces and welded parts shall be clean and free from any defects that will reduce the capability of the corridor to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies or deficient components or assemblies which are determined to be used "as is" shall not be used unless authorized by the contracting agency. External surfaces shall be free from burrs, slag, sharp edges and corners, exept where sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Individual inspection (see 4.6)

d. Daily inspection (see 4.7)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.5, 4.6, and 4.7.

4.4 First article inspection. Prior to beginning production of the joining corridor, the contractor shall fabricate one 4-ft x 8-ft honeycomb joining corridor panel and two joining corridors and subject them to the tests specified in Table I.

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TABLE I. Tests

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Sandwich panel shear	3.3.1.1.1	4.8.1	X		
Impact resistance	3.3.1.1.2	4.8.2	X		
Panel flatness	3.3.3	4.8.3	X	X	
Dimensional	3.6	4.8.4	X	X	X
Fire resistance	3.3.6	4.8.5	X		
Lifting from top	3.5.12	4.8.6	X		
Watertightness	3.5.1	4.8.7.1	X		
Humidity resistance	3.5.2	4.8.7.2	X		
Temperature range	3.5.2	4.8.7.3	X		
Temperature shock	3.5.2.1	4.8.7.4	X		
Heat transfer	3.5.2.2	4.8.7.5	X		
Door load	3.5.4	4.8.7.6	X		
Roof load, uniform	3.5.5	4.8.7.7	X		
Roof load, concentrated	3.5.5	4.8.7.8	X		
Blowing sand	3.5.6	4.8.7.9	X		
Sunshine	3.5.7	4.8.7.10	X		
Solar load	3.5.8	4.8.7.11	X		
Marine atmosphere	3.5.9	4.8.7.12	X		
Drop shock	3.5.10.1	4.8.8.1	X		
Rotational drop	3.5.10.2	4.8.8.2	X		
Blackout	3.5.11	4.8.9.1	X	X	X

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4.5 Quality conformance inspection. From each shipping lot as defined in the contract, a single joining corridor shall be chosen at random by the Government and subjected to the tests specified in Table I. Failure of any test shall be cause for inspection of the entire lot at the contractor's expense for similar defects. Defective joining corridors shall not be delivered without correction of the deficiency and then only with the permission of the contracting officer.

4.6 Individual inspection. Each joining corridor shall be subjected to the inspections and tests specified in Table I. Failure to meet the requirements of the drawings and this specification shall be cause for rejection. No joining corridor shall be reworked and retested without the expressed permission of the contracting agency.

4.7 Daily inspections. During the manufacture of the honeycomb panels, daily inspections shall be performed (see 4.8.10).

4.8 Inspections and tests.

4.8.1 Sandwich panel shear. Five preproduction sandwich panel shear specimens shall be subjected to MIL-STD-401, Test Method 5.2.4.1. The specimens shall be deemed acceptable if a shear stress in excess of 80 psi (551,584 N/m^2) in the T_L direction parallel to the core ribbons is achieved. The contracting agency reserves the right to require testing of production panels to the same criteria.

4.8.2 Impact resistance. A 24-in by 24-in (0.600-m by 0.600-m) sample of each exterior corridor panel, roof and floor shall be tested in accordance with MIL-STD-907, PAR 5.3.14.1, except for 2.5 # density honeycomb which shall be tested using a height of 16 inches (0.406M).

4.8.3 Panel flatness. The joining corridor shall be tested in accordance with MIL-STD-907, PAR 5.3.4.1.

4.8.4 Dimensional. The length (L), width (W), height (H) shall be measured and recorded for each corridor.

4.8.5 Fire resistance. Two 12-in by 12-in (300-mm by 300-mm) specimens of a production sandwich panel shall be subjected to tests in MIL-STD-907, PAR 5.3.1.1.

4.8.6 Lifting from top. The corridor under test shall be uniformly loaded to 2.5 times its rated gross weight (1,650 lbs (7,340 Newtons)). The corridor shall then be carefully lifted from all four top corners in such a way that no significant acceleration or deceleration forces are applied. The corridor shall be suspended by the lifting mechanism for 5 minutes and

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then carefully lowered to the ground. Upon completion of the test, the corridor shall show neither permanent deformation nor abnormality which will render it unsuitable for use.

4.8.7 Environmental tests

4.8.7.1 Watertightness. The joining corridor shall be subjected to watertightness test (panels) in accordance with MIL-STD-907, PAR 5.3.16.1.

4.8.7.2 Humidity resistance. The joining corridor shall be subjected to MIL-STD-810, Method 507.1, Procedure II, excluding steps 2-4, 7, and 8. After cycling, the corridor shall be examined and shall show no evidence of delamination, cracking, corrosion or deterioration of any part of the joining corridor.

4.8.7.3 Temperature range. The joining corridor shall be cold soaked in a mechanically refrigerated cold chamber at -70°F (-56.7°C) for a minimum of 16 hours. The chamber temperature shall then be raised to -65°F (-53.9°C). The corridor doors shall then be opened and closed. Any malfunction shall be recorded and reported. The joining corridor shall then be visually examined both internally and externally for material degradation. There shall be no damage to seals or other components and all hardware shall operate as specified during and after completion of this test.

4.8.7.4 Temperature shock. A representative corridor panel specimen measuring 4 ft (1.22 m) by 8 ft (2.44 m) shall be subjected to MIL-STD-810, Method 503.1. There shall be no separation, cracks, delamination or degradation of physical properties after completion of this test.

4.8.7.5 Heat transfer. The joining corridor shall be subjected to heat transfer test in accordance with MIL-STD-907. The overall heat transfer coefficient shall be determined and the requirements of 3.5.2.2 shall be satisfied.

4.8.7.6 Door loads. The standard personnel door on the corridor under test shall be subjected to tests specified in MIL-STD-907 for static and wind gust loading. Any evidence of unbonded components, damage to hardware, or improper open door operation shall constitute failure of this test.

4.8.7.7 Roof load, uniform. The joining corridor under test shall be placed on the ground and leveled. The roof of the corridor shall then be loaded uniformly with a weight of 3,153 lbs (14,025 Newtons) (40 lbs/square foot) and allowed to remain loaded for a period of 5 minutes. Upon completion of the test, the joining corridor shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.4.

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4.8.7.8 Roof load, concentrated. A load of 660 lbs (300 kg) shall be uniformly distributed over an area of 12 in by 24 in (300 mm by 600 mm) located at the weakest area of the roof of the joining corridor (centered on the roof). This load shall be allowed to remain there for a period of 5 minutes. Upon completion of the test, the corridor shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the tests in 4.8.4.

4.8.7.9 Blowing sand. The joining corridor shall be subjected to blowing sand test in accordance with MIL-STD-907. There shall be no damage to seals or other components and all hardware shall operate as specified herein upon completion of this test.

4.8.7.10 Sunshine. Representative samples of all gaskets and seals, as well as a 12-in by 12-in (300-mm by 300-mm) panel sample, shall be subjected to MIL-STD-810, Method 505.1, Procedure I. Upon completion of this test, the samples shall be examined and any evidence of degradation of physical properties shall constitute failure of this test.

4.8.7.11 Solar load. The joining corridor shall be subjected to solar load test-assembled shelter in accordance with MIL-STD-907, PAR 5.2.11.1. Upon completion of the solar load test, the roof panels shall be examined and any evidence of delamination or deformation will constitute failure.

4.8.7.12 Marine atmosphere. One representative sample of each fastener, jack, seal, panel, and other hardware which will be exposed to the atmosphere in the operation and storage modes shall be finished in accordance with the applicable drawings and subjected to tests as described in MIL-STD-810, Method 509.2, except that the salt solution shall have a concentrate of 5 percent and the exposure period shall be 96 hours. The test items shall show no evidence of corrosion upon completion of the test.

4.8.8 Shock tests

4.8.8.1 Drop shock. The joining corridor in its transport mode shall be lifted 12 in (300 mm) above a flat concrete surface. The distance between the concrete surface and each lower corner fitting of the corridor under test shall not vary more than 0.5 in (12.7 mm) at the time of the drop. The corridor shall then be allowed to free fall to the concrete surface. Upon completion of the test, the corridor shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be submitted to the test in 4.8.4.

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4.8.8.2 Rotational drop. The joining corridor shall be in its shipping mode (folded) with one edge supported 4 in (102 mm) above a flat concrete surface. The opposite edge shall be lifted a distance of 12 in (300 mm) above the concrete surface and allowed to free fall onto the surface. This procedure shall be repeated until each edge has been allowed to free fall. Upon completion of the test, the corridor shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.4.

4.8.9 Construction and assembly tests

4.8.9.1 Blackout. The joining corridor under test shall be assembled to a rigid shelter in accordance with MIL-S-29410, placed in an area of complete darkness, connected electrically to the appropriate power source and with the lights energized. With all apertures closed and latched, if applicable, all internal light fixtures operating, with the rigid shelter personnel door open and the joining corridor closed, no light shall be visible from outside the shelter or the joining corridor in any direction. This shall be verified by an observer from all sides including the top. If so equipped, the blackout curtain shall be placed in its track. The door shall be used to enter and exit the shelter. Any visible light shall be recorded.

4.8.10 Adhesive shear. A minimum of five sample adhesive shear specimens shall be prepared and tested daily as specified in 3.3.1.1.1 in accordance with ASTM E-865-82 procedure. The tests shall be conducted at normal temperature, 75° ± 5°F (23.9° ± 2.8°C). Tests shall be considered satisfactory if the shear stress developed exceeds 2,903 psi (20,015,600 N/m^2). Results of the tensile pull tests shall be recorded and identified as to the day of test and the lot from which the adhesive was taken. Failure of any specimen shall be cause for rejection of all panels bonded with that adhesive lot.

4.9 Test reports. The results of all testing shall be reported and written in compliance with MIL-STD-831 fully describing and illustrating the test procedures and test results.

5. PACKAGING

5.1 Preservation. The joining corridor in its shipping configuration requires no additional preservation.

5.2 Packing. As the corridor assembly in its shipping configuration is an exterior shipping container, no protective media need be installed prior to shipment. Conventional methods of blocking, bracing and strapping are to be employed during shipment by rail, road and air as described in MIL-STD-1186. Removable markings shall be affixed in accordance with MIL-STD-129 to show cube, weight, and destination specified by the contracting agency.

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5.3 Marking. In addition to the marking required by 3.9, each shelter assembly shall be marked for shipment in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The joining corridor covered by this specification is used to join one, two, three or four shelters together for use by air and ground U.S. Fleet Marine Forces to house and support tactical military equipment and operations in the field.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. First article testing, when required (see 3.1.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirement Title	Applicable DD No.	Option
4.9	Test procedures and reports	DI-T-4909	N/A

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the procurement contracting officer.

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6.4 Quality conformance. When a quality conformance inspection is required, the items to be tested should be a single joining corridor from each shipping lot. No lot should be accepted until the quality conformance tests have been successfully completed as required by 4.5.

6.5 Subject Term (Keyword) Listing

Characteristics, Dimensional
Complexing
Honeycomb
Joining Corridor
Panels, Sandwich
Workmanship

6.6 International standardization agreements. Certain provisions of this specification (3.1) are the subject of international standardization agreement ISO/DIS 668 and are regulated by the Coast Guard. When amendment, revision or cancellation of the specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy-MC
Project No. 5411-N039

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-29413B(MC)	Shelter, Joining Corridor For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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